Leakage, Shift and the Operating Point: A Two-Dataset Evaluation of Blood-Smear Leukemia Classifiers

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***Abstract*—Deep-learning models for leukemia detection in blood smears often report 97–100% accuracy, yet few are used in practice. We measured how much of such performance survives on new patients in two public datasets: 106 patients with acute myeloid leukemia (34 with acute promyelocytic leukemia, APL), split by diagnosis date, and C-NMC 2019 (73 subjects, acute lymphoblastic leukemia vs. normal B cells) with a held-out subject fold. The same models reached 88–95% accuracy under image-level cross-validation but 69–90% on new patients. Splitting the AUROC gap into its parts, patient leakage accounted for 0.04–0.11 when cells carried the patient's diagnosis and 0.01–0.03 when they carried their own label, and dataset shift for a further 0.17–0.24 on the later APL cohort and 0.02–0.06 on C-NMC. Leaky training did not produce worse rankers but a misleading operating point: thresholds tuned under leaky folds recovered only 32–66% of APL cells. Scan time was confounded with APL in the development cohort. A pre-specified pipeline—Macenko stain normalization, applied only after a label-free shift test, a fine-tuned network and engineered features combined in an ensemble—detected all 12 APL patients in the later cohort (AUROC 0.81, specificity 0.17) and 16 of 17 leukemic subjects in C-NMC (AUROC 0.96, specificity 0.79); it did not rank better than its best single member. A low-complexity variant with principal component analysis stored 20 times less per cell at a cost of 0.06 AUROC and 0.03 F1 score. A new laboratory would need about 20 confirmed positive cases to re-establish the threshold. Such models are suited to prioritizing smears for urgent review, not to ruling out APL.**

***Index Terms*—Acute promyelocytic leukemia, acute lymphoblastic leukemia, blood smear, data leakage, dataset shift, deep learning, external validation, multiple instance learning, sensitivity.**

I. Introduction

ACUTE leukemia is usually first suspected on a peripheral blood smear [1]. Acute promyelocytic leukemia (APL) shows how much rides on that first look: patients can bleed to death within days unless all-trans retinoic acid is started as soon as APL is suspected, often before genetic confirmation arrives. Reading smears takes experience that is not available everywhere, which is why automated smear analysis has attracted so much work.

Many deep-learning models report 97–100% accuracy for leukemia detection from single-cell images [2], [3], and weakly supervised models reach AUCs above 0.95 for APL [4], [5]. Yet few of these systems are used in practice. Part of the gap may lie in how they are evaluated. When cells from the same patient appear in both training and test sets, a model can succeed by recognizing the patient—the stain, the smear, the day it was scanned—rather than the disease; when the label is the patient's diagnosis copied onto every cell, this shortcut is always available [6], [7]. And accuracy alone hides the error that matters most clinically: a sick patient called healthy. Similar problems have been documented across machine-learning-based science [8] and medical imaging [9]–[11].

We measure these effects on a public cohort of 106 patients with acute myeloid leukemia (AML), 34 with APL, and repeat the analysis on the C-NMC 2019 dataset of acute lymphoblastic leukemia (ALL). Our contributions are:

- C1 *Leakage separated from shift.* With identical data and models, we compare image-level cross-validation (CV), patient-grouped CV and an external set, and split the gap into a leakage part and a shift part, for labels that belong to the patient and labels that belong to the cell; we also identify a scan-time confounder.
- C2 *Two kinds of external test.* A temporal split of the APL-AML cohort and a held-out subject fold of C-NMC, complemented by pooled CV over all 106 APL-AML patients.
- C3 *Sensitivity-first reporting.* Thresholds are fixed on development data for a recall of at least 0.95; we report missed patients, specificity, F1 score, negative predictive value (NPV), calibration and net benefit, and how many local cases a new laboratory needs to restore the operating point.
- C4 *Design decisions driven by the data.* An exploratory analysis and a label-free shift test decide the preprocessing; engineered, frozen and fine-tuned features, principal component analysis (PCA) and early stopping are each reported before and after.
- C5 *Pre-specified pipelines with their cost.* A primary and a low-complexity secondary pipeline, both trainable on a CPU, are reported with storage, model size and time per patient.

II. Related Work

Table I summarizes representative studies and the open question each leaves, linked to the contribution of this paper that addresses it.

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Cell-level classification. The C-NMC 2019 challenge [12] held out whole subjects for testing; ResNeXt models reached a weighted F1 score of 0.889 [13] and a five-CNN ensemble 0.886 [14]. Transformer and hybrid designs report 0.97–1.00 accuracy on C-NMC and ALL-IDB [2], [3], [15], and CNN features with classical classifiers reach 0.88 on three classes [16]. Because the split unit is often not reported, and some studies combine sources in which class and source coincide, it remains open how much of this carries over to new patients. A subject-disjoint benchmark on C-NMC found an AUROC of 0.913 [17], for a single cell-level label.

Patient-level diagnosis. Expert-level single-cell classification [18], [19] enabled multiple instance learning (MIL) for AML subtypes [20] and mutation status [21], [22]. Sidhom *et al.* [23] released the APL-AML cohort used here, validated their model on the later patients and compared it with ten clinicians; Manescu *et al.* [4] reported an AUC of 0.96 on the same patients under random three-fold CV, and Eckardt *et al.* detected APL in bone-marrow smears [5]. Engineered morphometry has held up in an external cohort for myelodysplasia [24]. The number of missed patients at a pre-set operating point, and calibration, are rarely reported.

Foundation models and shift. Hematology foundation models such as DinoBloom [25], built on DINOv2 [26], and their successors [27], [28] aim at robustness to scanners and stains; LeukemiaAttri images the same films with different microscopes [29], and a DinoBloom-based transformer trained on 6,610 individuals was validated externally, including on our cohort [30]. In histopathology, stain normalization and augmentation trade off differently across shifts [31].

III. Materials and Methods

A. Datasets

APL-AML. The public cohort of Sidhom *et al.* [23] contains single-cell images from Wright-stained peripheral blood smears of 106 adults with AML (34 APL, 72 other subtypes) treated at Johns Hopkins Hospital between 2010 and 2020 and captured with a CellaVision DM100 analyzer. The authors split it by date: 82 patients diagnosed before January 2019 form the Discovery cohort and 24 diagnosed later the Validation cohort. Of 25,724 images (about 360×363 pixels), 15,632 carry an operator-confirmed analyzer cell type (Table II; Fig. S1). Ten files (0.04%) are truncated and unreadable; they are kept as missing rows and imputed inside each training fold, and 54 are full microscope fields rather than single-cell crops. APL makes up 27% of Discovery and 50% of Validation patients, and the number of cells per patient is confounded with the diagnosis in opposite directions in the two cohorts. The original authors note that all Discovery smears scanned before 2018 were APL [23]; we examine this in Section IV-C.

C-NMC 2019. To test whether the findings hold in another laboratory, we used C-NMC 2019 [12], released by The Cancer Imaging Archive [32] and obtained from a Kaggle mirror whose file and subject counts match the official training set. It contains 10,661 segmented cells (450×450 pixels, black background) from 73 subjects: 47 with B-lineage ALL and 26 healthy individuals whose normal B cells serve as controls (Table S3). Before any analysis we designated official folds 0–1 as development set (42 subjects, 7,108 cells) and fold 2 as external set (31 subjects, 17 with ALL, 3,553 cells). Every cell carries its subject's diagnosis; unlike APL-AML, the ALL cells were selected as malignant lymphoblasts, so the label is also partly a property of the cell.

B. Learning Tasks

Five tasks are defined (Table III). In T1 every APL-AML cell inherits its patient's diagnosis, a common way of building single-cell datasets. In T2 the label is the cell's own type (blast versus other labeled white cells). T3 asks whether a patient has APL. T4 and T5 are the C-NMC counterparts of T1 and T3.

C. Preprocessing, Segmentation and Stain Normalization

Fig. 1 outlines the pipeline. A manifest records every image with its patient, cohort, diagnosis and cell label; no image is removed because of its label. For the engineered features, each image is resized to 192×192 pixels and separated into hematoxylin, eosin and residual channels by color deconvolution [33]. The nucleus is segmented by Otsu thresholding [34] of the smoothed hematoxylin channel, followed by morphological opening and selection of the component nearest the image center. Pale cytoplasm could not be separated reliably from neighboring red cells, so we describe it with two proxies: a basophilic region of blue-dominant pixels within one nuclear radius and a perinuclear ring 2–9 pixels wide (Fig. S5). For C-NMC, whose black background would be read as dense chromatin, pixels whose brightest channel is below 15 were set to white before engineered features were extracted; the CNN received the original images.

For the CNN we compared no preprocessing, edge sharpening (unsharp mask), contrast-limited adaptive histogram equalization (CLAHE), Reinhard color normalization [35] and Macenko stain normalization [36], with a rule fixed in advance: the method had to reduce the measurable cohort shift without lowering external performance. Macenko normalization estimates the two stain directions of each image in optical-density space and maps them, and their concentrations, onto reference stains estimated from 300 Discovery images. Because smear backgrounds are beige rather than white, the illumination of each image is taken from the 99th percentile of each color channel. Normalization is applied only when a label-free shift test is positive: a classifier assigns control-class cells (AML cells, normal B cells) to the development or external set with patient-grouped 5-fold CV, and a permutation test that shuffles the set labels between patients gives the p-value. The test reads the unlabeled external images but none of their diagnoses; it is transductive in the sense that a new site would run it on its own unlabeled images before deployment. Engineered features were always computed on the original images, because their color features are diagnostic in APL.

D. Feature Representations

The 93 engineered features describe nuclear shape (9), the basophilic region and nucleus-to-region ratio (10), stain and

TABLE I
REPRESENTATIVE PREVIOUS STUDIES (RESULTS AS REPORTED; NOT COMPARABLE ACROSS DATASETS) AND THE OPEN QUESTION EACH LEAVES, WITH THE CONTRIBUTION OF THIS PAPER (C1–C5) THAT ADDRESSES IT.

Study	Data	Method	Evaluation	Reported result	Open question (addressed by)
Prellberg & Kramer, 2019 [13]	C-NMC, ALL vs. normal cells	ResNeXt with squeeze-excitation	Challenge test set, unseen subjects	Weighted F1 0.889	Cell level only; missed cases and calibration not reported (C3, C5)
Mondal *et al.*, 2021 [14]	C-NMC	Ensemble of five CNNs	Preliminary test set, 28 subjects	F1 0.886; AUC 0.941	Cell level only; ensemble cost (C3, C5)
Kasim *et al.*, 2025 [16]	ALL-IDB2 + AML-LMU, 390 cells	CNN features + SVM/XGBoost	Image-level split	Accuracy 0.88 (3 classes)	Class and source coincide (C1, C2)
Jammal & Dahab, 2025 [2]	C-NMC	CNN–ViT, causal reasoning	Split unit not reported	Accuracy 1.00	Transfer to new patients unknown (C1, C2)
Jammal *et al.*, 2025 [3]	ALL image; C-NMC	Neuro-symbolic ViT	5-fold CV, split unit not reported	Accuracy 1.00; 0.97	Transfer to new patients unknown (C1, C2)
Albzour, 2026 [17]	C-NMC	ML, CNN and ViT benchmark	Subject-disjoint	AUROC 0.913	One cell-level label; leakage not separated from shift (C1, C5)
Hehr *et al.*, 2023 [20]	189 patients, 4 AML subtypes	Attention MIL	Patient-level CV	Subtype identification	Single laboratory; operating point not reported (C2, C3)
Sidhom *et al.*, 2021 [23]	APL-AML, 106 patients (*this cohort*)	Single-cell CNN + MIL	Temporal validation, reader study	AUC 0.890 → 0.743	Missed APL patients and calibration not reported (C3)
Manescu *et al.*, 2023 [4]	APL-AML (*this cohort*) and others	Annotation-free MIL (VGG-19)	Random 3-fold CV	AUC 0.96	Random rather than temporal split (C1, C2)
de Almeida *et al.*, 2023 [24]	352 + 63 slides	Engineered morphometry + MIL	CV and external cohort	Up to 0.90 AUC (MDS)	Not applied to acute leukemia (C4)
Koch *et al.*, 2024 [25]	13 datasets, 380k cells	DINOv2 foundation model	Linear probing, MIL	Outperforms generic encoders	Patient-level external evaluation open (C2, C5)
Dasdelen *et al.*, 2026 [30]	6,610 individuals; external incl. this cohort	DinoBloom + transformer MIL	Two external sets	Malignancy AUC 0.97	Needs large cohort and GPU (C5)

TABLE II
COMPOSITION OF THE APL-AML COHORT.

Cohort	Diagnosis	Patients	Cells	Labeled cells	Cells per patient median (range)
Discovery	AML	60	11,523	8,077	134 (98–984)
	APL	22	8,764	4,082	290 (112–1,066)
Validation	AML	12	3,589	1,893	252 (116–554)
	APL	12	1,848	1,580	140 (83–268)
Total		106	25,724	15,632	152 (83–1,066)

TABLE III
LEARNING TASKS (DEVELOPMENT / EXTERNAL SET). T1–T3: APL-AML (DISCOVERY / VALIDATION); T4, T5: C-NMC 2019 (FOLDS 0–1 / FOLD 2).

Task	Unit	Positive class	Size
T1 Diagnosis on cells	Cell	Cell from an APL patient (43.2%)	20,287 / 5,437 cells
T2 Blast detection	Cell	Labeled blast (28.0%)	9,694 / 2,863 cells
T3 Patient diagnosis	Patient	APL (22 of 82)	82 / 24 patients
T4 C-NMC cells	Cell	Cell from an ALL subject (67.7%)	7,108 / 3,553 cells
T5 C-NMC subjects	Subject	ALL (30 of 42)	42 / 31 subjects

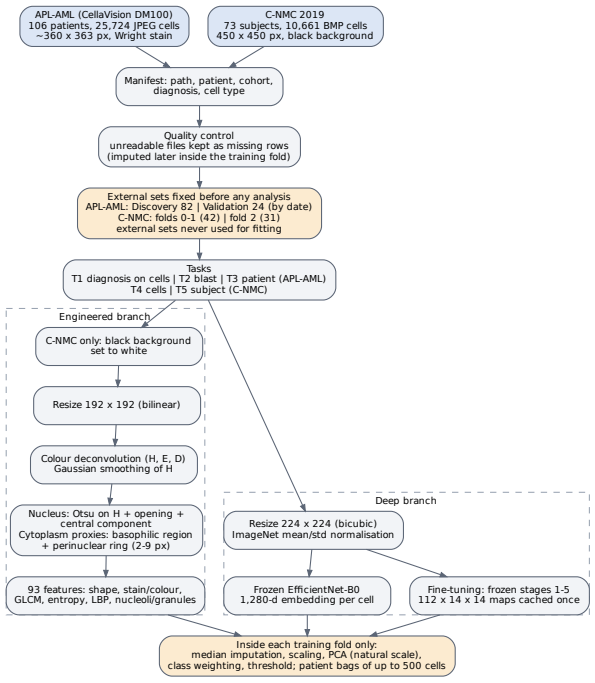


Fig. 1. Preprocessing for both datasets. Orange steps prevent leakage: the external sets are fixed before any analysis, and every statistic—imputation, scaling, PCA, class weights and the decision threshold—is fitted inside the training fold.

color in the nucleus and the ring (26), chromatin texture from gray-level co-occurrence matrices [37] and nuclear entropy (16), uniform local binary patterns [38] (28), and pale nuclear spots and dark granules (4). As a deep representation we used a frozen EfficientNet-B0 [39] pretrained on ImageNet (1,280 dimensions per cell); the fusion representation concatenates both after column-wise standardization. Because a frozen generic encoder may understate deep learning, we also fine-tuned EfficientNet-B0. Full fine-tuning was not feasible on our CPU-only workstation, so we froze the stem and first five stages, cached their output, and trained the last two stages, the head and a new classifier with AdamW [40] (learning rate 3×10^{-4}, cosine schedule, 6 epochs, best epoch on 15% of the training patients, up to 5,000 cells per fold, one seed; Table S1). DinoBloom [25] is supported by the released code but could not be evaluated because its weights could not be downloaded in our computing environment.

On the frozen embeddings we tested PCA, fitted on the training cells of each fold. The embeddings are standardized and projected onto the leading k eigenvectors $\mathbf{W}_k$ of their covariance [41], $\mathbf{z} = \mathbf{W}_k^\top \tilde{\mathbf{x}}$. We pre-specified $k \in \{16, 32, 64\}$ and added $k \in \{4, 8, 128, 256\}$ as a sensitivity analysis. A patient vector holds five statistics per dimension, so PCA reduces it from 6,400 to $5k$ values. The pooled component scores are centered and divided by one common scale ("natural scale"), so that the components keep their relative variances; standardizing each component separately would whiten them and give noise directions the same weight as the leading ones. Our first analysis used whitening; every PCA result in this paper uses the natural scale unless labeled otherwise. Further feature-engineering steps (correlation filter, per-slide stain normalization of features, L1 selection, a predicted differential count) are described in Table S1 and reported in Table S5.

E. Classification Models

Fig. S3 shows how the components fit together. At cell level we used L2-regularized logistic regression (LR) and LightGBM [42] with class-balanced weights, each as a single pipeline that imputes and scales with training data only. For patients and subjects (T3, T5), up to 500 randomly sampled cells represent each patient. The pooled model summarizes every feature by its mean, standard deviation and 10th, 50th and 90th percentiles and applies LR. Gated attention MIL [43] maps each cell to $\mathbf{h}_k = \mathrm{ReLU}(\mathbf{W}\mathbf{x}_k)$ (128 units), weights the cells by

$$a_k = \frac{\exp\{\mathbf{w}^\top(\tanh(\mathbf{V}\mathbf{h}_k) \odot \sigma(\mathbf{U}\mathbf{h}_k))\}}{\sum_j \exp\{\mathbf{w}^\top(\tanh(\mathbf{V}\mathbf{h}_j) \odot \sigma(\mathbf{U}\mathbf{h}_j))\}}, \qquad (1)$$

and classifies $\mathbf{z} = \sum_k a_k \mathbf{h}_k$; because the weights sum to one, the output does not depend on the number of cells. It is trained for 40 epochs with AdamW (learning rate 2×10^{-4}, weight decay 10^{-3}), class-weighted cross-entropy and random removal of 20% of cells. The fine-tuned CNN gives a patient score as the mean of its cell probabilities.

F. Final Pipelines and Pre-Specification

Before the final analysis we fixed one primary and one secondary pipeline (Fig. 2) and committed the specification to the code repository (commit `f47f3ec`, 21 October 2026, 07:19 UTC+3). The primary pipeline normalizes stains when the shift test is positive, scores every cell with the fine-tuned CNN, and averages the percentile scores of three patient-level members—the mean fine-tuned cell probability, engineered features with pooled LR, and engineered features with attention MIL—with no weights fitted. The secondary pipeline was declared as a low-complexity option: stain normalization, the frozen CNN, PCA with 64 components, pooled statistics and LR. Both use a threshold fixed on development out-of-fold scores for a recall of at least 0.95. Their intended use is to prioritize smears for urgent review and confirmatory testing, not to rule APL out.

What the pre-specification does and does not protect. The APL-AML Validation cohort was never used to fit a model, threshold or calibration. It was, however, not a single-use test set. In our first analysis (20 October 2026) and during this revision we evaluated many configurations on it—feature-engineering steps, PCA sizes, preprocessing variants on a subset—and those numbers were known when later choices were made, including the pre-specification. The primary pipeline was assembled from the components that had been most consistent across both datasets, but we cannot exclude that knowledge of Validation results shaped it. The APL-AML Validation numbers should therefore be read as those of a reused test set; the C-NMC external fold (also evaluated for several configurations, but never used for selection), the pooled analysis and the decomposition into leakage and shift provide the more protected evidence.

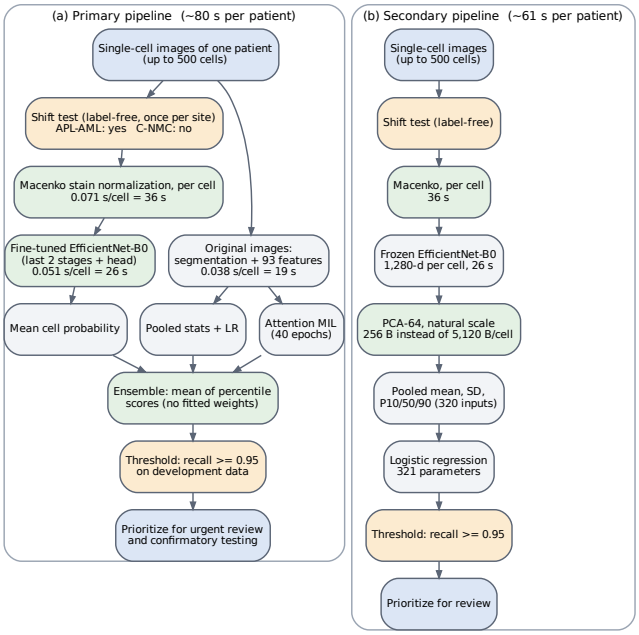


Fig. 2. End-to-end pipelines from the single-cell images of a patient to the decision, with the setting and measured cost of each step (500 cells per patient, 2 vCPUs). Orange: steps fixed on development data. Green: components retained because they improved performance (primary) or reduced complexity (secondary).

G. Evaluation Protocol and Metrics

Every cell-level model was evaluated in three ways (Fig. S4): 5-fold CV on the development set with image-level folds, as is common in the literature; 5-fold CV with patient-grouped folds; and the external set. We define *leakage* as image-level minus patient-grouped CV AUROC (same data, only the split differs) and *shift* as patient-grouped CV minus external AUROC (new patients). We also scored the external set with the models trained under image-level folds, to see whether leaky training harms the model or only the estimate. Within each training fold, an inner 80/20 split of the same kind fitted a calibration temperature [44]. External predictions of patient-level models average all fold models over three seeds.

Since missing an APL patient can delay life-saving treatment, whereas a false alarm leads to a confirmatory genetic test, every decision threshold was fixed in advance on devel-

opment out-of-fold scores to reach a recall of at least 0.95. At that operating point we report recall with the number of missed positives, specificity, F1 score, NPV and accuracy; AUROC as a threshold-free summary; and calibration as the Brier score, calibration intercept and slope [45] and expected calibration error (ECE). Confidence intervals (95%) come from bootstrap resampling of patients [46]; differences between models use a paired patient-level bootstrap, with Holm correction [47] when several comparisons share a reference. C-NMC was analyzed with the same protocol.

Three further analyses address what the external tests alone cannot. First, because 24 external patients give wide intervals [48], [49], we repeated the main patient-level comparisons with three repeats of 5-fold CV over all 106 APL-AML patients, stratified by diagnosis and cohort, with thresholds chosen by inner CV. Second, we simulated local recalibration: k confirmed local patients ($k/2$ APL, $k = 2, \ldots, 8$) are drawn from the Validation cohort, the threshold is lowered if needed to just below the lowest local APL score, and the remaining $24 - k$ patients are scored, over 2,000 draws. Third, decision-curve analysis [50] estimated the net benefit at an APL prevalence of 10%, after logistic recalibration on Discovery and a shift to the target prevalence.

H. Implementation

All experiments ran on a CPU-only workstation (2 vCPUs, 7.8 GB RAM; Table S2). Table S1 lists every setting; settings were fixed in advance or taken from library defaults, and none was tuned on an external set. Reporting follows TRIPOD+AI [51] (checklist in Table S14).

IV. Results

A. Exploratory Analysis and the Shift Test

Both datasets are imbalanced in different ways: APL makes up 27% of Discovery and 50% of Validation patients; ALL makes up 71% of C-NMC development and 55% of external subjects. We therefore used class-balanced weights, did not resample cells, set thresholds for the positive class, and report recall, specificity, missed cases, F1 score and AUROC rather than accuracy alone.

Fig. 3 summarizes the APL-AML exploratory analysis. Every feature group shifted between cohorts, stain and color most (median standardized mean difference 0.34). Among AML patients, a classifier separated Validation from Discovery cells with an AUROC of 0.753 on stain and color features (permutation $p = 0.005$), 0.704 on all engineered features ($p = 0.010$) and 0.583 on frozen CNN embeddings ($p = 0.039$; Table S4). Thirty-seven feature pairs were nearly redundant ($|r| > 0.9$), and single-feature patient-level AUROCs in the two cohorts correlated only moderately ($r = 0.38$). C-NMC looked different (Fig. S2): feature shifts were small (median 0.04–0.16 per group) and normal B cells could not be assigned to their set (AUROC 0.44–0.47, $p \geq 0.45$ for every representation). By the pre-stated rule, stain normalization was therefore applied to APL-AML only. On the full data, CNN embeddings of normalized images remained about as identifiable as before (0.603, $p = 0.06$), whereas on the pilot subset used for the selection normalization had lowered the identifiability from 0.575 to 0.543; Macenko normalization reduces color differences but does not remove every acquisition signature the CNN can see.

B. Leakage Versus Shift

Table IV separates the two sources of optimism. When each cell carried its patient's diagnosis (T1), image-level CV reported AUROCs of 0.938–0.987, patient-grouped CV 0.867–0.901 and the external cohort 0.658–0.734. Leakage accounted for 0.04–0.11 of the gap and shift for 0.17–0.24. When the label described the cell itself (T2), leakage was 0.01–0.03 and shift nearly zero. On C-NMC cells (T4), leakage (0.07–0.08) was larger than shift (0.02–0.06), consistent with a random subject-disjoint fold of one collection. Leakage is thus largest when the label belongs to the patient, as expected, but on the temporally split APL-AML cohort the shift between periods was the larger part of the overstatement.

Leaky training did not produce worse models. Classifiers trained under image-level folds ranked external cells as well as or slightly better than those trained under grouped folds (T1: 0.677–0.760 vs. 0.658–0.734), presumably because each fold model saw cells of every patient. What leakage corrupted was the operating point: a threshold tuned for a recall of 0.95 on leaky folds recovered only 32–66% of the cells of new APL patients, against 69–81% for thresholds tuned on grouped folds. Table V shows the result in the terms most studies report: the same models reached 0.88–0.95 accuracy under image-level CV at a 0.5 threshold, but 0.69–0.73 on new APL-AML patients, 0.87–0.90 for blasts and 0.82–0.83 on C-NMC.

C. Scan Time as a Confounder

The images carry no scan dates, but the analyzer numbers the cells it signs sequentially, so the median cell number of a patient is a time proxy. It separated Validation from Discovery patients almost perfectly (AUROC 0.992), which supports its validity. Within Discovery, it predicted APL with an AUROC of 0.775 (Table VI). In the Validation cohort it carried no diagnostic information (0.479), so the confounding is confined to the development data. The images themselves record scan time: among AML patients only, pooled features separated early from late scans with AUROCs of 0.74–0.79 for every representation, including shape and texture (0.767); Macenko normalization lowered this for color features from 0.760 to 0.658 but did not remove it. To see whether the diagnostic models depend on it, we repeated the Discovery CV in the 31 patients (18 APL) scanned while both diagnoses were being scanned, where scan time no longer predicted APL (0.325). Engineered features kept their performance (CV AUROC 0.927 on all patients, 0.929 in the window), whereas the frozen CNN dropped from 0.957 to 0.858, stain and color features from 0.881 to 0.805, and engineered features computed on normalized images from 0.869 to 0.729. In every case the model score added information beyond scan time (likelihood-ratio $p < 0.001$), but scores and scan time were correlated (Spearman 0.23–0.55). Part of the high development

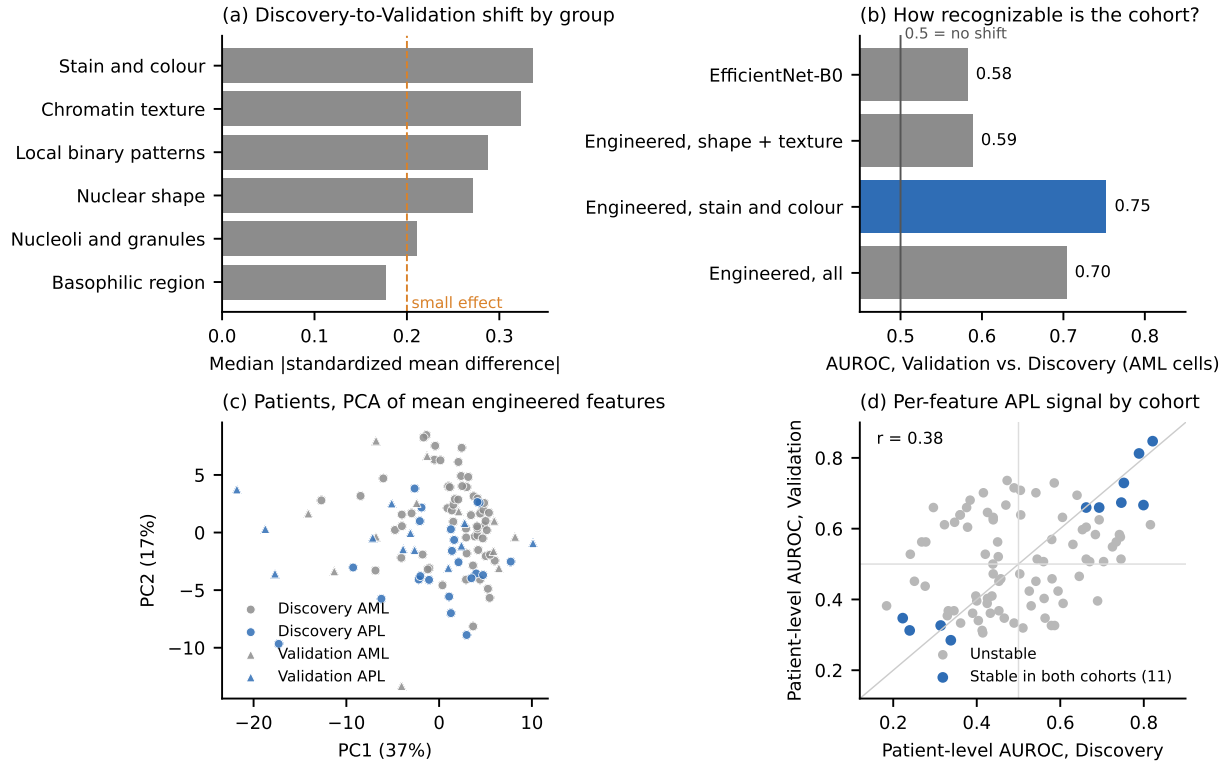


Fig. 3. Exploratory analysis of APL-AML. (a) Shift of engineered feature groups from Discovery to Validation. (b) AUROC of a classifier that tells the two cohorts apart from AML cells; 0.5 means no detectable shift. (c) Patients on the first two principal components of their mean engineered features. (d) Patient-level AUROC of each feature in the two cohorts; blue points separate the classes consistently.

TABLE IV

HOW MUCH OF THE GAP BETWEEN REPORTED AND REAL PERFORMANCE IS DUE TO LEAKAGE AND HOW MUCH TO DATASET SHIFT (CELL LEVEL, MEAN OVER SEEDS; ± SD ACROSS SEEDS). LEAKAGE = IMAGE-LEVEL CV AUROC − PATIENT-GROUPED CV AUROC (SAME DEVELOPMENT DATA, ONLY THE SPLIT DIFFERS). SHIFT = PATIENT-GROUPED CV AUROC − EXTERNAL AUROC (NEW PATIENTS; TEMPORAL FOR APL-AML, A HELD-OUT SUBJECT FOLD FOR C-NMC). LEAKY-TRAINED EXTERNAL: EXTERNAL AUROC OF THE MODELS TRAINED UNDER IMAGE-LEVEL FOLDS. RECALL, SPECIFICITY AND F1 ON THE EXTERNAL SET AT THE OPERATING POINT FIXED ON DEVELOPMENT DATA.

		AUROC				Decomposition of the gap		External, fixed operating point		
Task	Model	Image CV	Grouped CV	External [95% CI]	Leaky-trained ext.	Leakage	Shift	Recall	Spec.	F1
T1 APL-AML: diagnosis on cells (patient label)	Engineered + LightGBM	0.987	0.896	0.669 [0.495, 0.811]	0.723	+0.091 ± 0.011	+0.227 ± 0.041	0.69	0.55	0.54
	Engineered + LR	0.938	0.895	0.658 [0.478, 0.798]	0.677	+0.042 ± 0.006	+0.237 ± 0.017	0.73	0.46	0.52
	EfficientNet-B0 + LR	0.975	0.867	0.694 [0.556, 0.811]	0.717	+0.108 ± 0.009	+0.173 ± 0.039	0.81	0.43	0.56
	Fusion + LR	0.985	0.901	**0.734 [0.595, 0.848]**	0.760	+0.083 ± 0.009	+0.167 ± 0.031	0.78	0.54	0.59
T2 APL-AML: blast detection (cell label)	Engineered + LightGBM	0.978	0.954	**0.957 [0.931, 0.977]**	0.956	+0.024 ± 0.003	-0.003 ± 0.003	0.99	0.62	0.58
	Engineered + LR	0.965	0.951	0.951 [0.922, 0.972]	0.951	+0.014 ± 0.003	-0.000 ± 0.003	0.98	0.70	0.63
	EfficientNet-B0 + LR	0.956	0.927	0.920 [0.874, 0.956]	0.920	+0.029 ± 0.002	+0.007 ± 0.006	0.96	0.64	0.58
	Fusion + LR	0.964	0.940	0.932 [0.894, 0.961]	0.932	+0.024 ± 0.003	+0.009 ± 0.007	0.96	0.68	0.61
T4 C-NMC: cells (subject label)	Engineered + LR	0.933	0.864	**0.840 [0.730, 0.902]**	0.848	+0.069 ± 0.008	+0.023 ± 0.025	0.95	0.58	0.89
	EfficientNet-B0 + LR	0.912	0.834	0.775 [0.640, 0.859]	0.786	+0.079 ± 0.014	+0.059 ± 0.023	0.96	0.43	0.87
	Fusion + LR	0.939	0.864	0.817 [0.692, 0.892]	0.825	+0.075 ± 0.009	+0.047 ± 0.026	0.95	0.54	0.88

performance of CNN and color features therefore rests on time-linked acquisition differences, which would show up as "shift" when the model meets a later cohort; the 31-patient window is small, and these estimates are imprecise.

D. Choosing the Preprocessing

Table VII compares preprocessing options on a fixed subset of up to 60 cells per patient. Edge sharpening raised development performance (cells 0.873 to 0.887) but lowered external performance (cells 0.661 to 0.631; patients 0.667 to 0.639), widening the gap. CLAHE changed little (external 0.667 for cells and patients) and did not reduce the cohort shift (0.575 to 0.581). Reinhard normalization removed most of the shift (0.524) but lowered external cell AUROC (0.644). Macenko normalization reduced the shift (0.543) and was the only option that raised external performance at both levels (cells 0.690, patients 0.694), so it met the rule. Estimating stain vectors once per patient was faster but lowered patient-level development CV by 0.017 and failed the pre-specified rule, so normalization is done per cell (0.071 s per cell). On C-NMC, where no shift was detected, Macenko normalization lowered external AUROC slightly (subjects 0.882 to 0.857), supporting conditional use. These differences are small compared with the uncertainty of 24 external patients, and the subset included

TABLE V
THE SAME MODELS EVALUATED THE WAY MANY STUDIES REPORT THEM (IMAGE-LEVEL 5-FOLD CV, ACCURACY AT A 0.5 THRESHOLD) AND ON PATIENTS THEY HAVE NEVER SEEN (EXTERNAL SET). MEAN OVER SEEDS.

		Image-level CV (leaky)			External set (new patients)			
Task	Model	Accuracy	AUROC	F1	Accuracy	AUROC	Recall	F1
T1 APL-AML, diagnosis on cells	Engineered + LightGBM	0.950	0.987	0.942	0.691	0.669	0.245	0.342
	Fusion + LR	0.942	0.985	0.933	0.715	0.734	0.370	0.466
	Fine-tuned EfficientNet-B0	–	–	–	0.732	0.773	0.340	0.464
T2 APL-AML, blast detection	Engineered + LightGBM	0.928	0.978	0.875	0.898	0.957	0.895	0.786
	Fusion + LR	0.911	0.964	0.845	0.866	0.932	0.850	0.725
	Fine-tuned EfficientNet-B0	–	–	–	0.904	0.960	0.883	0.792
T4 C-NMC, cells	Engineered + LR	0.877	0.933	0.907	0.827	0.840	0.896	0.877
	Fusion + LR	0.894	0.939	0.921	0.816	0.817	0.895	0.871
	Fine-tuned EfficientNet-B0	–	–	–	0.825	0.841	0.946	0.882

TABLE VI
SCAN-TIME CONFOUNDING IN THE APL-AML DISCOVERY COHORT (82 PATIENTS). TIME PROXY: MEDIAN ANALYZER CELL NUMBER OF A PATIENT'S SIGNED CELLS. DIAGNOSIS MODELS: POOLED-STATISTICS LR, PATIENT-LEVEL 5-FOLD CV $\times$ 3 SEEDS, ON ALL PATIENTS AND ON THE PATIENTS SCANNED WHILE BOTH DIAGNOSES WERE BEING SCANNED (TIME WINDOW). SCORE ADDS TO TIME: LIKELIHOOD-RATIO TEST OF THE OUT-OF-FOLD SCORE ADDED TO A LOGISTIC MODEL ON THE TIME RANK. TIME FROM FEATURES: AUROC FOR EARLY VS. LATE SCAN AMONG AML PATIENTS ONLY.

Representation	CV AUROC, all	CV AUROC, time window	Score adds to time, p	Time from features (AML)
Engineered (all)	0.927	0.929	<0.001	0.794
Engineered: stain and colour	0.881	0.805	<0.001	0.760
Engineered: shape + texture	0.904	0.887	<0.001	0.767
EfficientNet-B0 (frozen)	0.957	0.858	<0.001	0.743
Engineered, on Macenko images	0.869	0.729	<0.001	–
Time proxy alone: Validation vs. Discovery 0.992; APL vs. AML in Discovery 0.775; within the window 0.325 (31 patients, 18 APL).				

Validation patients, as disclosed above.

E. Engineered, Frozen and Fine-Tuned Features

For blast detection on new patients, engineered features with LightGBM reached an AUROC of 0.957 (95% CI 0.931–0.977) against 0.920 for frozen EfficientNet-B0 embeddings (paired difference $+0.034$, 0.008–0.065), and on C-NMC cells 0.840 against 0.775 (Table VIII). Fine-tuning closed this gap: external AUROC rose to 0.773 for T1 (frozen 0.694), 0.960 for blasts and 0.841 on C-NMC cells. For blasts it also gave a better operating point (specificity 0.84 instead of 0.62 at recall 0.95, F1 0.74 instead of 0.58). For T1, the threshold chosen on Discovery flagged almost every Validation cell (recall 0.996, specificity 0.17), the operating-point problem seen throughout the temporal cohort.

F. Dimensionality Reduction: Before and After PCA

Table IX and Fig. 4 show the effect of PCA on the pooled patient model (natural scale; full sweep in Table S6). Without reduction, each patient is described by 6,400 values, about 100 per APL-AML training patient.

On APL-AML without stain normalization, PCA raised external AUROC from 0.708 to 0.792 with 64 components but lowered the decisions at the clinical threshold (6 instead of 8 of 12 APL patients detected; F1 0.57 instead of 0.67). On C-NMC it cost a little on every metric (AUROC 0.908 without, 0.853 with 64 components; 16 instead of 17 ALL subjects detected). In pooled CV over all 106 APL-AML patients, 32 and 64 components ranked as well as the unreduced embedding (ΔAUROC $+0.000$ and $+0.006$; Table S9), and 16 components were worse (-0.057). The pre-declared rule (highest development CV among $k \in \{16, 32, 64\}$) selected 64 on both datasets.

PCA and stain normalization interacted. After Macenko normalization, PCA-64 no longer improved ranking: external AUROC fell from 0.701 to 0.639 and F1 from 0.79 to 0.76, with recall unchanged (11 of 12). Normalization had already removed part of the cohort-specific color variation that, without it, PCA removed together with some signal. PCA keeps directions of large variance, not those that separate the classes, and the external disease signal grew steadily with the number of components on both datasets (Fig. 4c). We therefore use PCA only to reduce complexity (Section IV-H), not as a way to gain accuracy.

Whitening was a pitfall: with each component standardized, development CV rose to 0.99 while external AUROC on C-NMC fell from 0.849 (8 components) to 0.655 (256), the signature of a classifier fitting noise directions (dashed lines in Fig. 4).

G. Training Settings: Epochs and Early Stopping

We compared the fixed 40 epochs of attention MIL with a data-driven alternative: inside every training fold, nested 3-fold patient-level CV chose the epoch with the lowest validation loss. This did not help (Table S7). External AUROC changed by less than 0.01 on APL-AML and fell by 0.02–0.06 on C-NMC, and F1 and calibration did not improve. Each inner model was trained on about 35 and validated on about 17 APL-AML patients (C-NMC: 18 and 9), and the chosen epoch ranged from 1 to 40 for the same model, so early stopping made results less stable across seeds. We kept the fixed schedule. Learning curves (Fig. S7, computed with whitened PCA inputs from our first analysis and after the configuration was fixed) show that held-out AUROC plateaued after about 10 epochs while held-out loss rose, i.e., the fixed schedule cost no ranking but produced overconfident probabilities.

TABLE VII

PREPROCESSING BEFORE FEATURE EXTRACTION (FROZEN EFFICIENTNET-B0, UP TO 60 CELLS PER PATIENT, SAME FOLDS AND CLASSIFIERS FOR EVERY VARIANT). COHORT SHIFT: AUROC FOR ASSIGNING CONTROL-CLASS CELLS TO THEIR COHORT (0.5 = NO DETECTABLE SHIFT). GAP: DEVELOPMENT CV MINUS EXTERNAL AUROC. THE RULE FIXED IN ADVANCE SELECTED THE METHOD THAT REDUCED THE SHIFT WITHOUT LOWERING EXTERNAL PERFORMANCE. C-NMC, WHICH SHOWED NO SHIFT, IS GIVEN FOR COMPARISON.

		Cells (T1)			Patients (T3)		
Preprocessing	Cohort shift	Dev. CV (SD; worst fold)	External	Gap	Dev. CV (SD; worst fold)	External	Gap
APL-AML (6,360 cells, 106 patients)							
None	0.575	0.873 (0.036; 0.818)	0.661	0.212	0.956 (0.047; 0.875)	0.667	0.289
Edge sharpening (unsharp mask)	0.561	0.887 (0.030; 0.834)	0.631	0.256	0.963 (0.050; 0.875)	0.639	0.324
Contrast enhancement (CLAHE)	0.581	0.861 (0.024; 0.827)	0.667	0.194	0.939 (0.057; 0.833)	0.667	0.273
Reinhard colour normalization	0.524	0.873 (0.026; 0.836)	0.644	0.228	0.972 (0.039; 0.896)	0.681	0.291
Macenko stain normalization (selected)	0.543	0.850 (0.037; 0.785)	0.690	0.160	0.942 (0.052; 0.875)	0.694	0.247
Macenko, stain vectors per patient	0.554	0.859 (0.048; 0.775)	0.615	0.244	0.925 (0.070; 0.833)	0.611	0.314
C-NMC (3,773 cells, 73 subjects; no shift detected)							
None	0.549	0.832 (0.050; 0.753)	0.746	0.086	0.889 (0.222; 0.444)	0.882	0.007
Macenko stain normalization	0.560	0.818 (0.058; 0.708)	0.729	0.088	0.861 (0.238; 0.389)	0.857	0.004

TABLE VIII

ENGINEERED, FROZEN AND FINE-TUNED FEATURES AT CELL LEVEL ON BOTH DATASETS. THE FINE-TUNED MODEL ADAPTS THE LAST TWO STAGES OF EFFICIENTNET-B0 (TABLE S1; ONE SEED); THE OTHER MODELS ARE AVERAGED OVER THREE SEEDS. LEAKAGE: IMAGE-LEVEL MINUS PATIENT-GROUPED CV AUROC; SHIFT: GROUPED CV MINUS EXTERNAL AUROC. RECALL, SPECIFICITY AND F1 ON THE EXTERNAL SET AT THE OPERATING POINT FIXED ON DEVELOPMENT DATA. BEST EXTERNAL AUROC PER TASK IN BOLD.

Task	Model	Image CV	Grouped CV	External [95% CI]	Leakage	Shift	Recall	Spec.	F1
T1 APL-AML, diagnosis on cells	Engineered + LightGBM	0.987	0.896	0.669 [0.495, 0.811]	+0.091	+0.227	0.69	0.55	0.54
	Engineered + LR	0.938	0.895	0.658 [0.478, 0.798]	+0.042	+0.237	0.73	0.46	0.52
	EfficientNet-B0 + LR	0.975	0.867	0.694 [0.556, 0.811]	+0.108	+0.173	0.81	0.43	0.56
	Fusion + LR	0.985	0.901	0.734 [0.595, 0.848]	+0.083	+0.167	0.78	0.54	0.59
	Fine-tuned EfficientNet-B0	–	0.875	**0.773 [0.612, 0.890]**	–	+0.101	1.00	0.17	0.55
T2 APL-AML, blast detection	Engineered + LightGBM	0.978	0.954	0.957 [0.931, 0.977]	+0.024	-0.003	0.99	0.62	0.58
	Engineered + LR	0.965	0.951	0.951 [0.922, 0.972]	+0.014	-0.000	0.98	0.70	0.63
	EfficientNet-B0 + LR	0.956	0.927	0.920 [0.874, 0.956]	+0.029	+0.007	0.96	0.64	0.58
	Fusion + LR	0.964	0.940	0.932 [0.894, 0.961]	+0.024	+0.009	0.96	0.68	0.61
	Fine-tuned EfficientNet-B0	–	0.968	**0.960 [0.928, 0.981]**	–	+0.008	0.95	0.83	0.74
T4 C-NMC, cells	Engineered + LR	0.933	0.864	0.840 [0.730, 0.902]	+0.069	+0.023	0.95	0.58	0.89
	EfficientNet-B0 + LR	0.912	0.834	0.775 [0.640, 0.859]	+0.079	+0.059	0.96	0.43	0.87
	Fusion + LR	0.939	0.864	0.817 [0.692, 0.892]	+0.075	+0.047	0.95	0.54	0.88
	Fine-tuned EfficientNet-B0	–	0.853	**0.841 [0.715, 0.917]**	–	+0.012	0.99	0.42	0.88

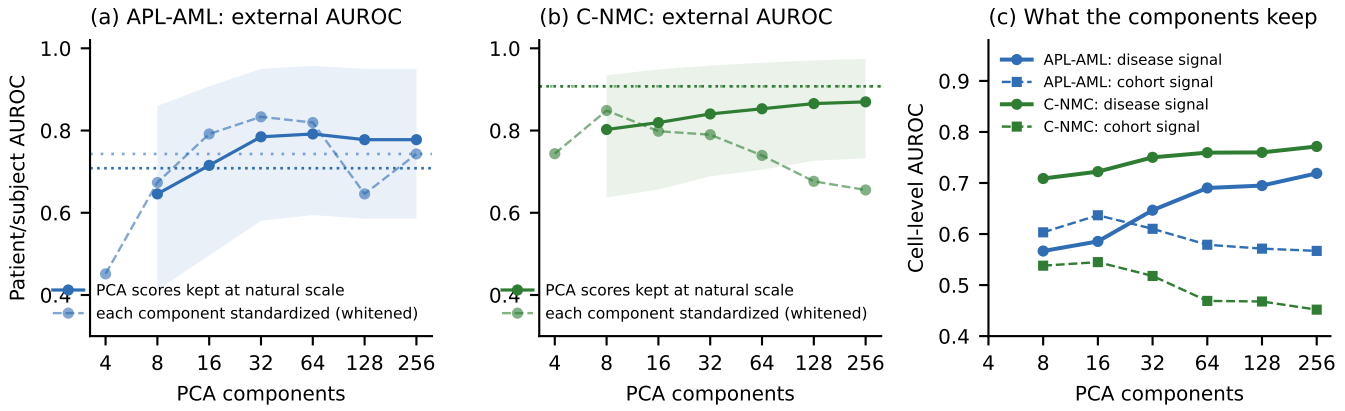


Fig. 4. PCA of EfficientNet-B0 embeddings, pooled patient/subject model. (a) APL-AML and (b) C-NMC: external AUROC versus the number of components at natural scale (solid; band: 95% CI) and whitened (dashed); horizontal lines: no PCA. (c) External cell-level disease signal (circles) and cohort signal for control-class cells (squares; 0.5 = none).

Replacing average pooling of the fine-tuned CNN by attention pooling over spatial positions changed development AUROC by -0.020, $+0.014$ and $+0.005$ on three folds and was not pursued.

TABLE IX
IMPACT OF PCA ON PATIENT/SUBJECT-LEVEL DETECTION (EXTERNAL SETS). BEFORE: UNREDUCED 1,280-DIMENSIONAL EMBEDDING. AFTER: PCA FITTED ON DEVELOPMENT CELLS, SCORES KEPT AT THEIR NATURAL SCALE. RULE: NUMBER OF COMPONENTS WITH THE HIGHEST DEVELOPMENT CV AUROC (PRE-DECLARED). THRESHOLD METRICS AT THE OPERATING POINT FIXED ON DEVELOPMENT DATA.

Dataset	Representation	AUROC	ΔAUROC	Acc.	Recall (missed)	Spec.	F1	NPV
APL-AML	Before: no PCA	0.708	–	0.67	0.67 (4/12)	0.67	0.67	0.67
	After: PCA-32	0.785	+0.076	0.62	0.50 (6/12)	0.75	0.57	0.60
	After: PCA-64	0.792	+0.083	0.62	0.50 (6/12)	0.75	0.57	0.60
	After: PCA-256 (rule)	0.778	+0.069	0.62	0.50 (6/12)	0.75	0.57	0.60
C-NMC	Before: no PCA	0.908	–	0.84	1.00 (0/17)	0.64	0.87	1.00
	After: PCA-32	0.840	-0.067	0.71	0.94 (1/17)	0.43	0.78	0.86
	After: PCA-64	0.853	-0.055	0.74	0.94 (1/17)	0.50	0.80	0.88
	No PCA (rule)	0.908	–	0.84	1.00 (0/17)	0.64	0.87	1.00

H. Final Pipelines on New Patients

Table X and Fig. 5 trace how each component changed performance on new patients; Table XI sets the pipelines against their cost.

Primary pipeline. On APL-AML, the primary pipeline detected all 12 APL patients in the Validation cohort (recall 1.00, NPV 1.00; AUROC 0.806, 95% CI 0.615–0.956), but flagged 10 of 12 AML patients (specificity 0.17, F1 0.71). On C-NMC (no normalization, because no shift was detected) it reached an AUROC of 0.960 and detected 16 of 17 ALL subjects at a specificity of 0.79 (F1 0.89). The ensemble did not rank better than its best member (APL-AML: attention MIL 0.812; C-NMC: fine-tuned CNN 0.962), but it gave the most balanced operating point: on APL-AML it matched the pooled engineered model (no missed patient, specificity 0.17) where attention MIL flagged every AML patient and the fine-tuned CNN missed two APL patients; on C-NMC it kept the specificity of the pooled model (0.79) where the fine-tuned CNN reached 0.57. Its value is robustness of the operating point, not better ranking.

What each step contributed. Fine-tuning was the most consistent improvement at cell level on both datasets. Stain normalization helped the frozen CNN at the operating point (1 instead of 4 missed APL patients; F1 0.79 instead of 0.67) but not the fine-tuned CNN (AUROC 0.771 vs. 0.778), which appears to absorb much of the color shift itself.

Secondary pipeline and the cost of PCA. With PCA-64, the patient-level classifier has 321 instead of 6,401 parameters and stores 256 instead of 5,120 bytes per cell; it trains twice as fast and, for attention MIL, with 7 times fewer parameters. The price on APL-AML was 0.06 AUROC (0.701 to 0.639), 0.03 F1 (0.79 to 0.76) and 0.08 specificity, with the same recall. Image-level steps dominate the time per patient—Macenko normalization 0.071 s, the CNN 0.051 s and engineered features 0.038 s per cell—so the primary pipeline needs about 80 s per 500-cell patient on our CPU, 45% of it for normalization, and PCA mainly saves storage, memory and retraining effort rather than time per diagnosis.

I. Replication on C-NMC

Table XII repeats the experiments on C-NMC. All six patient-level models ranked the 31 external subjects well (AUROC 0.853–0.941), engineered features best (pooled LR 0.941). Unlike on APL-AML, the development operating point transferred: every model detected at least 15 of the 17 ALL subjects, at specificities up to 0.79. C-NMC external ALL subjects contributed more cells than normal subjects (median 161 vs. 41; count alone AUROC 0.735), whereas in the development folds the count carried no signal (0.453). The model scores were nearly uncorrelated with the count ($|\rho| \leq 0.29$), remained discriminative after regressing it out (AUROC 0.89–0.95), and added information beyond it ($p < 0.001$; Table S12), so the number of cells does not explain the C-NMC results.

J. Calibration

Patient-level probabilities were poorly calibrated on both external sets (Table S11, Fig. S9). On APL-AML, calibration intercepts were positive (1.0–1.3 for raw outputs), i.e., APL risk was underestimated, as expected when the APL prevalence rises from 27% to 50%; on C-NMC they were negative (-0.7 to -1.6) as the ALL prevalence fell from 71% to 55%. Slopes varied widely (0.3–5.4), and the Brier score of the primary pipeline after Platt scaling on development data was 0.231 on APL-AML and 0.136 on C-NMC. With 24 and 31 external patients, these estimates are imprecise, but they show that predicted probabilities should not be used as risks without local recalibration.

K. Pooled Analysis and Local Recalibration

In pooled CV over all 106 APL-AML patients (Table S9), the unreduced EfficientNet-B0 embedding ranked patients best (AUROC 0.953), better than engineered features with pooled statistics (-0.059, significant after Holm correction) and the promyelocyte fraction used as a manual biomarker (0.677). Thresholds chosen inside each training fold transferred to the held-out folds (recall 0.89–1.00), which supports the view that the failure on the Validation cohort reflects temporal shift; because patients from both periods were mixed, this analysis does not measure transfer to a later cohort.

Setting a new site's threshold just below the lowest score of $m = k/2$ local APL cases gave the same expected recall for every model, about $m/(m+1)$ (Fig. 6a; Table S10). This follows from exchangeability and does not depend on the model, so about 19 confirmed local APL cases are needed before such a threshold can be expected to reach a recall of 0.95. Lowering the development threshold only when a local APL case would otherwise be missed changed little for models whose development threshold already reached a recall of 0.92–1.00 on the Validation cohort; it mainly cost specificity (frozen CNN with Macenko: 0.58 to 0.42 with eight local cases), and raised the recall of the fine-tuned CNN only from 0.83 to 0.86 (Fig. 6b). For the models with recall below 1.00, the probability of missing at least one APL patient among the remaining 16 stayed at 0.66–0.68. At an APL prevalence of 10% (Fig. 6c), the primary pipeline gave the highest net benefit at a 10% threshold probability (0.050 against 0 for testing everyone and 0.017–0.042 for the other models), and benefits close to testing everyone below 5%.

TABLE X

WHAT EACH ADDITION CHANGES ON NEW PATIENTS (EXTERNAL SETS; CELLS: T1/T4, PATIENTS: T3/T5). EACH BLOCK STARTS FROM A FROZEN IMAGENET CNN; Δ: AUROC CHANGE RELATIVE TO THE FIRST ROW OF THE BLOCK. THRESHOLD METRICS AT THE OPERATING POINT FIXED ON DEVELOPMENT DATA (TARGET RECALL 0.95). CELL ROWS: MEAN OF THREE SEEDS (FROZEN) OR ONE SEED (FINE-TUNED). THE ENSEMBLE AVERAGES PERCENTILE SCORES OF THE FINE-TUNED CNN (WITH MACENKO ON APL-AML), ENGINEERED POOLED LR AND ENGINEERED ATTENTION MIL.

Dataset	Level	Model	AUROC [95% CI]	ΔAUROC	Accuracy	Recall	Missed	Specificity	F1
APL-AML	Cell	Frozen CNN + LR	0.694 [0.556, 0.811]	–	0.56	0.81	–	0.43	0.56
		Frozen CNN + engineered (fusion)	0.734 [0.595, 0.848]	+0.040	0.62	0.78	–	0.54	0.59
		+ fine-tuning	0.773 [0.612, 0.890]	+0.079	0.45	1.00	–	0.17	0.55
		+ fine-tuning + Macenko	0.752 [0.601, 0.864]	+0.058	0.44	0.99	–	0.15	0.55
APL-AML	Patient	Frozen CNN, pooled + LR	0.708 [0.493, 0.907]	–	0.67	0.67	4/12	0.67	0.67
		+ PCA-64	0.792 [0.594, 0.957]	+0.083	0.62	0.50	6/12	0.75	0.57
		Frozen CNN + Macenko	0.701 [0.482, 0.907]	-0.007	0.75	0.92	1/12	0.58	0.79
		Frozen CNN + Macenko + PCA-64 (secondary)	0.639 [0.400, 0.867]	-0.069	0.71	0.92	1/12	0.50	0.76
		Fine-tuned CNN, mean of cells	0.778 [0.576, 0.938]	+0.069	0.75	0.83	2/12	0.67	0.77
		Fine-tuned CNN + Macenko	0.771 [0.563, 0.941]	+0.062	0.67	0.83	2/12	0.50	0.71
		Engineered, pooled + LR	0.757 [0.555, 0.938]	+0.049	0.58	1.00	0/12	0.17	0.71
		Engineered, attention MIL	0.812 [0.614, 0.963]	+0.104	0.50	1.00	0/12	0.00	0.67
		Primary: ensemble (Macenko fine-tuned + engineered)	0.806 [0.615, 0.956]	+0.097	0.58	1.00	0/12	0.17	0.71
C-NMC	Cell	Frozen CNN + LR	0.775 [0.640, 0.859]	–	0.79	0.96	–	0.43	0.87
		Frozen CNN + engineered (fusion)	0.817 [0.692, 0.892]	+0.042	0.83	0.95	–	0.54	0.88
		+ fine-tuning	0.841 [0.715, 0.917]	+0.067	0.81	0.99	–	0.42	0.88
C-NMC	Patient	Frozen CNN, pooled + LR	0.908 [0.786, 0.992]	–	0.84	1.00	0/17	0.64	0.87
		+ PCA-64	0.853 [0.705, 0.965]	-0.055	0.74	0.94	1/17	0.50	0.80
		Fine-tuned CNN, mean of cells	0.962 [0.887, 1.000]	+0.055	0.81	1.00	0/17	0.57	0.85
		Engineered, pooled + LR	0.941 [0.840, 1.000]	+0.034	0.87	0.94	1/17	0.79	0.89
		Engineered, attention MIL	0.912 [0.791, 1.000]	+0.004	0.77	0.88	2/17	0.64	0.81
		Primary: ensemble of the three members above	0.960 [0.886, 1.000]	+0.053	0.87	0.94	1/17	0.79	0.89

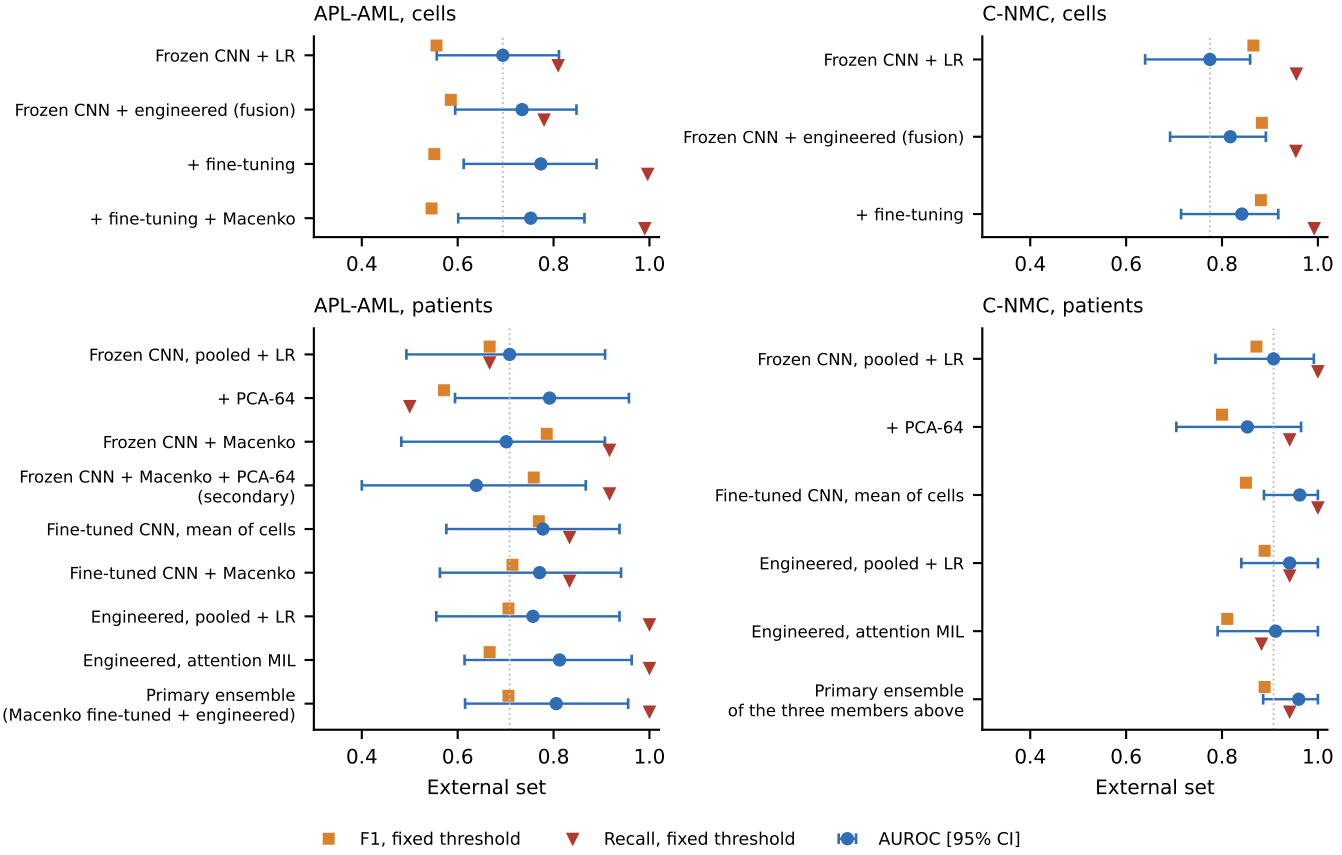


Fig. 5. Effect of each added component on new patients (external sets): AUROC with 95% CI (circles), and F1 score (squares) and recall (triangles) at the operating point fixed on development data. Dotted line: AUROC of the first row.

L. Comparison With Existing Methods on the Same Cohort

Table XIII compares our pipelines with existing methods on the same patients. Sidhom *et al.* reported an AUC of 0.743

TABLE XI

PERFORMANCE AND COMPLEXITY ON APL-AML (24 EXTERNAL PATIENTS). TIME PER PATIENT ASSUMES 500 CELLS ON OUR CPU (2 VCPUS): MACENKO NORMALIZATION 0.071 S, CNN FORWARD PASS 0.051 S AND ENGINEERED FEATURES 0.038 S PER CELL; CLASSIFIER TIMES ARE MEASURED. STORAGE: BYTES KEPT PER CELL AFTER FEATURE EXTRACTION. PARAMETERS AND TRAINING TIME REFER TO THE PATIENT-LEVEL CLASSIFIER (INCLUDING PCA FITTING).

	Performance (external, fixed threshold)							Complexity			
Pipeline	AUROC	Acc.	Recall	Missed	Spec.	F1	NPV	Storage/cell	Parameters	Training (s)	Time/patient (s)
Primary: Macenko + fine-tuned CNN + engineered, ensemble	0.806	0.58	1.00	0/12	0.17	0.71	1.00	5,492 B	4,038,033	631.00	80
Macenko + frozen CNN, no PCA	0.701	0.75	0.92	1/12	0.58	0.79	0.88	5,120 B	6,401	0.26	61
Secondary: Macenko + frozen CNN + PCA-64	0.639	0.71	0.92	1/12	0.50	0.76	0.86	256 B	321	6.15	61

TABLE XII

REPLICATION ON THE SECOND DATASET, C-NMC 2019 (ALL VS. NORMAL B CELLS; DEVELOPMENT: OFFICIAL FOLDS 0–1, 42 SUBJECTS; EXTERNAL: FOLD 2, 31 UNSEEN SUBJECTS, 17 WITH ALL). SAME FEATURES, MODELS, PROTOCOL AND OPERATING-POINT RULE AS FOR APL-AML. LEAKAGE: IMAGE-LEVEL MINUS SUBJECT-GROUPED CV AUROC; SHIFT: GROUPED CV MINUS EXTERNAL AUROC.

(a) Cell level: every cell carries its subject's diagnosis (mean of 3 seeds)

Model	Image CV	Grouped CV	External [95% CI]	Recall	Spec.	F1	Leakage	Shift
Engineered + LR	0.933	0.864	0.840 [0.730, 0.902]	0.95	0.58	0.89	+0.069	+0.023
EfficientNet-B0 + LR	0.912	0.834	0.775 [0.640, 0.859]	0.96	0.43	0.87	+0.079	+0.059
Fusion + LR	0.939	0.864	0.817 [0.692, 0.892]	0.95	0.54	0.88	+0.075	+0.047

(b) Subject level: is this subject's sample leukemic?

Representation	Aggregation	CV AUROC	External [95% CI]	Recall (missed)	Spec.	F1	NPV
Engineered	Pooled + LR	0.906	0.941 [0.840, 1.000]	0.94 (1/17)	0.79	0.89	0.92
EfficientNet-B0	Pooled + LR	0.855	0.920 [0.807, 0.992]	0.94 (1/17)	0.71	0.86	0.91
Fusion	Pooled + LR	0.878	0.929 [0.824, 0.996]	1.00 (0/17)	0.71	0.89	1.00
Engineered	Attention MIL	0.881	0.912 [0.791, 1.000]	0.88 (2/17)	0.64	0.81	0.82
EfficientNet-B0	Attention MIL	0.874	0.853 [0.704, 0.963]	1.00 (0/17)	0.43	0.81	1.00
Fusion	Attention MIL	0.887	0.853 [0.696, 0.967]	1.00 (0/17)	0.57	0.85	1.00

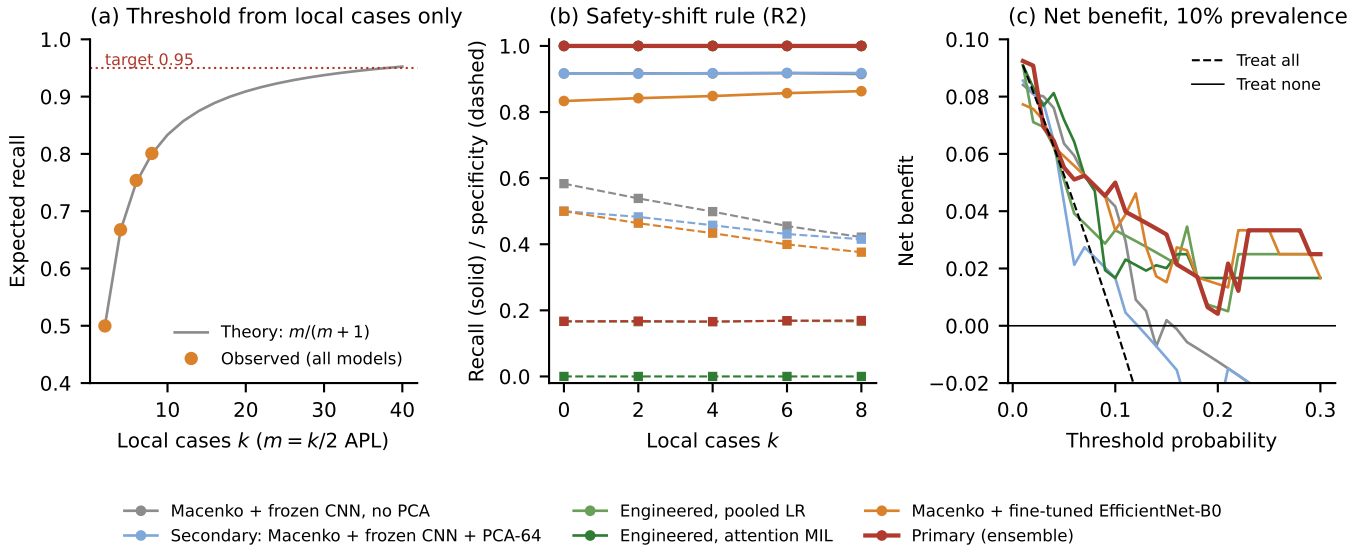


Fig. 6. Restoring the operating point at a new site (APL-AML Validation cohort). (a) Expected recall when the threshold is set just below the lowest score of $m = k/2$ local APL cases, observed and theoretical $m/(m+1)$. (b) Recall (solid) and specificity (dashed) when the development threshold is lowered only as far as needed to catch the local APL cases. (c) Decision curves at 10% APL prevalence; dashed: test every patient; horizontal: test none.

in the later cohort [23] and compared their model with ten clinicians on the same 24 patients, finding it equivalent or better (reported only graphically); Manescu *et al.* reported 0.96 with random three-fold CV [4]. Neither reported results at a fixed recall. Re-run on the identical split with our operating-point rule, the promyelocyte proportion reached an AUROC of 0.611 and the full differential count 0.681; averaging single-cell predictions per patient [23] reached 0.681–0.799 and the CNN-feature-plus-SVM design of [16] 0.806. The primary pipeline reached 0.806 and missed no APL patient. With 24 external patients the confidence intervals of all learned models overlap; the clearest differences are over the manual

biomarkers, in the number of missed patients, and between random CV (0.96) and temporal validation.

V. Discussion

Four findings stand out. First, how a model is evaluated changes the reported number more than how it is built, and the overstatement has two separable sources. Patient leakage inflated AUROC by up to 0.11 when cells carried their patient's diagnosis, but much less when the label described the cell; the shift to a later cohort added up to 0.24. Leaky training did not make the models worse rankers, but it made their thresholds wrong. A study that reports image-level CV therefore overstates both the ranking and, more seriously, the recall that can be expected in practice. This held on two datasets and may explain why subject-disjoint challenge entries report F1 scores near 0.89 rather than near 1.0 [13], [14].

Second, the temporal APL-AML split carries a confounder that a patient-grouped CV within the development cohort cannot remove: APL patients were scanned earlier than most AML patients. Restricting the Discovery cohort to the period in which both diagnoses were scanned left engineered morphology features unaffected but cost the frozen CNN 0.10 AUROC, which suggests that generic deep features partly learn when a smear was scanned, and that the large shift component on the temporal cohort is partly the unwinding of this confounding rather than a change in disease appearance. Studies on this cohort, ours included, should report how their results depend on scan time.

Third, the data decided which design choices helped. A label-free test for whether cells can be traced to their set found a stain shift in APL-AML and none in C-NMC, and this one observation explained several results: stain normalization helped on APL-AML and slightly hurt on C-NMC, development thresholds transferred to C-NMC but not to the later APL cohort, and PCA interacted with normalization. Fine-tuning was the only modelling choice that helped on both datasets; nested early stopping and attention pooling did not improve on simpler settings, and PCA traded a little performance for a much smaller model.

Fourth, ranking transferred better than decisions, and small external sets make honest reporting harder than it looks. On the later APL cohort the primary pipeline missed no APL patient but cleared only 2 of 12 AML patients, and every other model either missed patients or flagged most of the AML cohort. A threshold set from m local positives has an expected recall of only $m/(m+1)$, so about 20 confirmed local cases are needed before a recall of 0.95 can be expected, and 24 external patients are far below the sample sizes recommended for external validation [48], [49]. For a condition in which a missed diagnosis can be fatal, a model like ours is best used to prioritize smears for urgent review and testing, with its threshold confirmed on local cases.

Limitations. The two datasets differ in disease (APL vs. other AML; ALL vs. normal B cells), so the replication tests the evaluation principles rather than one clinical model across sites. The external sets are small, and the APL-AML Validation cohort was evaluated repeatedly during development and is best regarded as a reused test set (Section III-F). The scan-time proxy is indirect. Cell labels in APL-AML come from an analyzer verified by an operator, and our cytoplasm features rely on proxies. Fine-tuning was partial, limited to 5,000 cells per fold and run with one seed on a CPU; stain augmentation [31], full fine-tuning and hematology foundation models such as DinoBloom [25] may perform better, and an independent AML cohort with APL cases remains to be tested.

VI. Conclusion

Reported accuracy for leukemia detection from blood smears depends strongly on how the evaluation is set up. Across two public datasets, splitting cells instead of patients inflated performance and, above all, the expected recall, while a temporal shift and a scan-time confounder accounted for the larger part of the gap on the APL cohort. A pipeline built step by step from the data—stain normalization only where a shift was detected, fine-tuned and interpretable features, and pre-specified combination and thresholds—missed no APL patient in a later cohort and 1 of 17 leukemic subjects in a second dataset, at low specificity on the temporal cohort. We recommend that studies in this area split data by patient, separate leakage from shift, test for acquisition shift and confounding before choosing preprocessing, report recall, missed cases, specificity, F1 score, NPV and calibration at a pre-specified operating point, and report the computational cost of every component.

Data and Code Availability

The APL-AML cohort is available at https://github.com/sidhomj/DeepAPL and C-NMC 2019 from The Cancer Imaging Archive [32]. Code, split files, label maps, the pre-specification and per-model outputs will be released at [repository URL withheld for review].

References

[1] J. D. Khoury, E. Solary, O. Abla *et al.*, "The 5th edition of the World Health Organization classification of haematolymphoid tumours: Myeloid and histiocytic/dendritic neoplasms," *Leukemia*, vol. 36, pp. 1703–1719, 2022.

[2] F. Jammal and M. Dahab, "CausalFormer-HMC: A hybrid memory-driven transformer with causal reasoning and counterfactual explainability for leukemia diagnosis," *Front. Cell Dev. Biol.*, vol. 13, p. 1674393, 2025.

[3] F. Jammal, M. Dahab, and A. Y. Bayahya, "Neuro-Bridge-X: A neuro-symbolic vision transformer with meta-XAI for interpretable leukemia diagnosis from peripheral blood smears," *Diagnostics*, vol. 15, no. 16, p. 2040, 2025.

[4] P. Manescu, P. Narayanan, C. Bendkowski *et al.*, "Detection of acute promyelocytic leukemia in peripheral blood and bone marrow with annotation-free deep learning," *Sci. Rep.*, vol. 13, p. 2562, 2023.

[5] J.-N. Eckardt, T. Schmittmann, S. Riechert, M. Kramer, A. S. Sulaiman, K. Sockel, F. Kroschinsky, J. Schetelig, L. Wagenführ, U. Schuler *et al.*, "Deep learning identifies acute promyelocytic leukemia in bone marrow smears," *BMC Cancer*, vol. 22, p. 201, 2022.

[6] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.

[7] A. J. DeGrave, J. D. Janizek, and S.-I. Lee, "AI for radiographic COVID-19 detection selects shortcuts over signal," *Nature Machine Intelligence*, vol. 3, pp. 610–619, 2021.

[8] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.

TABLE XIII
COMPARISON WITH EXISTING METHODS ON THE SAME APL-AML COHORT. UPPER BLOCK: RESULTS AS REPORTED BY THE ORIGINAL STUDIES, WHOSE PROTOCOLS DIFFER FROM OURS. LOWER BLOCKS: METHODS RE-RUN ON THE IDENTICAL SPLIT (TRAINED ON DISCOVERY, TESTED ONCE ON THE 24 VALIDATION PATIENTS) WITH THE SAME OPERATING-POINT RULE (SENSITIVITY ≥ 0.95 ON DISCOVERY). THE PRIMARY PIPELINE IS COMPARED WITH EACH OF ITS MEMBERS.

Method	Evaluation	AUROC [95% CI]	Recall (missed)	Spec.	F1	NPV	Acc.
As reported in the literature							
Single-cell CNN, Sidhom *et al.* [23]	Temporal validation, cell level	0.743	–	–	–	–	–
Annotation-free MIL, Manescu *et al.* [4]	Random 3-fold CV, patient level	0.96 ± 0.02	–	–	–	–	–
Ten clinicians (reader study in [23])	Same 24 Validation patients, smear images	shown only graphically; the authors' MIL model was equivalent or better					
Re-run on the identical split (Validation cohort, patient level)							
Promyelocyte proportion (biomarker in [23])	Same protocol	0.611 [0.371, 0.837]	1.00 (0/12)	0.00	0.67	0.00	0.50
Differential count + LR (comparator in [30])	Same protocol	0.681 [0.437, 0.886]	1.00 (0/12)	0.08	0.69	1.00	0.54
CNN features + SVM (design of [16])	Same protocol	0.806 [0.615, 0.963]	0.50 (6/12)	0.75	0.57	0.60	0.62
Single-cell classifier, engineered (design of [23])	Same protocol	0.681 [0.445, 0.908]	0.75 (3/12)	0.50	0.67	0.67	0.62
Single-cell classifier, EfficientNet-B0 (design of [23])	Same protocol	0.799 [0.580, 0.972]	0.58 (5/12)	0.92	0.70	0.69	0.75
Single-cell classifier, fusion (design of [23])	Same protocol	0.764 [0.536, 0.957]	0.75 (3/12)	0.75	0.75	0.75	0.75
This work (pre-specified pipelines and their members)							
Primary pipeline (ensemble)	Same protocol	0.806 [0.615, 0.956]	1.00 (0/12)	0.17	0.71	1.00	0.58
Member: engineered, pooled LR	Same protocol	0.757 [0.555, 0.938]	1.00 (0/12)	0.17	0.71	1.00	0.58
Member: engineered, attention MIL	Same protocol	0.812 [0.614, 0.963]	1.00 (0/12)	0.00	0.67	0.00	0.50
Member: Macenko + fine-tuned CNN	Same protocol	0.771 [0.563, 0.941]	0.83 (2/12)	0.50	0.71	0.75	0.67
Secondary pipeline (Macenko + frozen CNN + PCA-64)	Same protocol	0.639 [0.400, 0.867]	0.92 (1/12)	0.50	0.76	0.86	0.71

[9] J. R. Zech, M. A. Badgeley, M. Liu, A. B. Costa, J. J. Titano, and E. K. Oermann, "Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study," *PLoS Med.*, vol. 15, no. 11, p. e1002683, 2018.

[10] M. Roberts, D. Driggs, M. Thorpe *et al.*, "Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans," *Nat. Mach. Intell.*, vol. 3, pp. 199–217, 2021.

[11] G. Varoquaux and V. Cheplygina, "Machine learning for medical imaging: methodological failures and recommendations for the future," *npj Digital Medicine*, vol. 5, p. 48, 2022.

[12] R. Gupta, S. Gehlot, and A. Gupta, "C-NMC: B-lineage acute lymphoblastic leukaemia: A blood cancer dataset," *Med. Eng. Phys.*, vol. 103, p. 103793, 2022.

[13] J. Prellberg and O. Kramer, "Acute lymphoblastic leukemia classification from microscopic images using convolutional neural networks," arXiv:1906.09020 (ISBI 2019 C-NMC Challenge), 2019.

[14] C. Mondal, M. K. Hasan, M. Ahmad, M. A. Awal, M. T. Jawad, A. Dutta, M. R. Islam, and M. A. Moni, "Ensemble of convolutional neural networks to diagnose acute lymphoblastic leukemia from microscopic images," *Inform. Med. Unlocked*, vol. 27, p. 100794, 2021.

[15] R. D. Labati, V. Piuri, and F. Scotti, "All-IDB: The acute lymphoblastic leukemia image database for image processing," in *Proc. 18th IEEE Int. Conf. Image Process. (ICIP)*, 2011, pp. 2045–2048.

[16] S. Kasim, S. Malek, J. Tang, X. N. Kiew, S. Cheen, B. Liew, N. Saidon, R. Ezman, and R. Shariff, "Multiclass leukemia cell classification using hybrid deep learning and machine learning with CNN-based feature extraction," *Sci. Rep.*, vol. 15, p. 23782, 2025.

[17] N. Albzour, "A leakage-aware comparative benchmark of machine learning, deep learning, and transformer models for reliable leukemia detection," arXiv:2606.24944, 2026.

[18] C. Matek, S. Schwarz, K. Spiekermann, and C. Marr, "Human-level recognition of blast cells in acute myeloid leukaemia with convolutional neural networks," *Nat. Mach. Intell.*, vol. 1, pp. 538–544, 2019.

[19] C. Matek, S. Krappe, C. Münzenmayer, T. Haferlach, and C. Marr, "Highly accurate differentiation of bone marrow cell morphologies using deep neural networks on a large image data set," *Blood*, vol. 138, no. 20, pp. 1917–1927, 2021.

[20] M. Hehr, A. Sadafi, C. Matek, P. Lienemann, C. Pohlkamp, T. Haferlach, K. Spiekermann, and C. Marr, "Explainable AI identifies diagnostic cells of genetic AML subtypes," *PLOS Digit. Health*, vol. 2, no. 3, p. e0000187, 2023.

[21] B.-H. Wei, X. C.-H. Tsai, K.-J. Sun, M.-Y. Lo, S.-Y. Hung, W.-C. Chou, H.-F. Tien, H.-A. Hou, and C.-Y. Chen, "Annotation-free deep learning for predicting gene mutations from whole slide images of acute myeloid leukemia," *npj Precis. Oncol.*, vol. 9, p. 35, 2025.

[22] J.-N. Eckardt, J. M. Middeke, S. Riechert, T. Schmittmann, A. S. Sulaiman, M. Kramer *et al.*, "Deep learning detects acute myeloid leukemia and predicts NPM1 mutation status from bone marrow smears," *Leukemia*, vol. 36, pp. 111–118, 2022.

[23] J.-W. Sidhom, I. J. Siddarthan, B.-S. Lai, A. Luo, B. C. Hambley, J. Bynum, A. S. Duffield, M. B. Streiff, A. R. Moliterno, P. Imus, C. B. Gocke, L. P. Gondek, A. E. DeZern, A. S. Baras, T. Kickler, M. J. Levis, and E. Shenderov, "Deep learning for diagnosis of acute promyelocytic leukemia via recognition of genomically imprinted morphologic features," *npj Precis. Oncol.*, vol. 5, p. 38, 2021.

[24] J. G. de Almeida, E. Gudgin, M. Besser *et al.*, "Computational analysis of peripheral blood smears detects disease-associated cytomorphologies," *Nat. Commun.*, vol. 14, p. 4378, 2023.

[25] V. Koch, S. J. Wagner, S. Kazeminia, E. Sancar, M. Hehr, J. A. Schnabel, T. Peng, and C. Marr, "DinoBloom: A foundation model for generalizable cell embeddings in hematology," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, 2024, pp. 520–530.

[26] M. Oquab, T. Darcet, T. Moutakanni *et al.*, "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, 2024.

[27] A. Muminov and A. Pham, "CytoDINO: Risk-aware and biologically-informed adaptation of DINOv3 for bone marrow cytomorphology," arXiv:2512.17930, 2025.

[28] C. Zamora, H. Zuniga, U. Orozco-Rosas, and K. Picos, "Retrieval-augmented vision foundation models for robust leukemia cell classification across multiple microscopy datasets," arXiv:2608.10657, 2026.

[29] A. Rehman, T. Meraj, A. M. Minhas, A. Imran, M. Ali, and W. Sultani, "A large-scale multi domain leukemia dataset for the white blood cells detection with morphological attributes for explainability," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, 2024.

[30] M. F. Dasdelen, I. Kukuljan, P. Lienemann, F. Ozlugedik, A. Sadafi, M. Hehr, K. Spiekermann, C. Pohlkamp, and C. Marr, "AI-based

hematological malignancy prediction from peripheral blood smears in a large diagnostic laboratory cohort," *Leukemia*, 2026.

[31] D. Tellez, G. Litjens, P. Bándi, W. Bulten, J.-M. Bokhorst, F. Ciompi, and J. van der Laak, "Quantifying the effects of data augmentation and stain color normalization in convolutional neural networks for computational pathology," *Medical Image Analysis*, vol. 58, p. 101544, 2019.

[32] S. Mourya, S. Kant, P. Kumar, A. Gupta, and R. Gupta, "ALL challenge dataset of ISBI 2019 (C-NMC 2019) [data set]," The Cancer Imaging Archive, 2019.

[33] A. C. Ruifrok and D. A. Johnston, "Quantification of histochemical staining by color deconvolution," *Anal. Quant. Cytol. Histol.*, vol. 23, no. 4, pp. 291–299, 2001.

[34] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern.*, vol. 9, no. 1, pp. 62–66, 1979.

[35] E. Reinhard, M. Ashikhmin, B. Gooch, and P. Shirley, "Color transfer between images," *IEEE Comput. Graph. Appl.*, vol. 21, no. 5, pp. 34–41, 2001.

[36] M. Macenko, M. Niethammer, J. S. Marron, D. Borland, J. T. Woosley, X. Guan, C. Schmitt, and N. E. Thomas, "A method for normalizing histology slides for quantitative analysis," in *Proc. IEEE Int. Symp. Biomed. Imaging (ISBI)*, 2009, pp. 1107–1110.

[37] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.

[38] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, 2002.

[39] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 6105–6114.

[40] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.

[41] I. T. Jolliffe and J. Cadima, "Principal component analysis: A review and recent developments," *Philos. Trans. R. Soc. A*, vol. 374, no. 2065, p. 20150202, 2016.

[42] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 3146–3154.

[43] M. Ilse, J. M. Tomczak, and M. Welling, "Attention-based deep multiple instance learning," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 2127–2136.

[44] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 1321–1330.

[45] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, "Calibration: the Achilles heel of predictive analytics," *BMC Medicine*, vol. 17, p. 230, 2019.

[46] B. Efron and R. Tibshirani, "Bootstrap methods for standard errors, confidence intervals, and other measures of statistical accuracy," *Stat. Sci.*, vol. 1, no. 1, pp. 54–75, 1986.

[47] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

[48] R. D. Riley, T. P. A. Debray, G. S. Collins, L. Archer, J. Ensor, M. van Smeden, and K. I. E. Snell, "Minimum sample size for external validation of a clinical prediction model with a binary outcome," *Statistics in Medicine*, vol. 40, no. 19, pp. 4230–4251, 2021.

[49] G. S. Collins, E. O. Ogundimu, and D. G. Altman, "Sample size considerations for the external validation of a multivariable prognostic model: a resampling study," *Statistics in Medicine*, vol. 35, no. 2, pp. 214–226, 2016.

[50] A. J. Vickers and E. B. Elkin, "Decision curve analysis: A novel method for evaluating prediction models," *Med. Decis. Making*, vol. 26, no. 6, pp. 565–574, 2006.

[51] G. S. Collins, K. G. M. Moons, P. Dhiman *et al.*, "TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, p. e078378, 2024.